



deti

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Introduction to Computer Graphics

Sonic Hub World

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Links

Site



<https://duartebranco.github.io/sonic-hub-world/>

Repo



<https://github.com/duartebranco/sonic-hub-world>

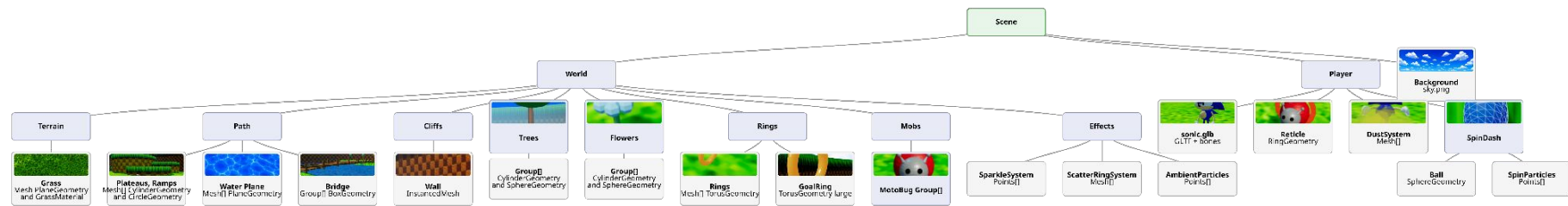
Sonic Hub World

Objective:

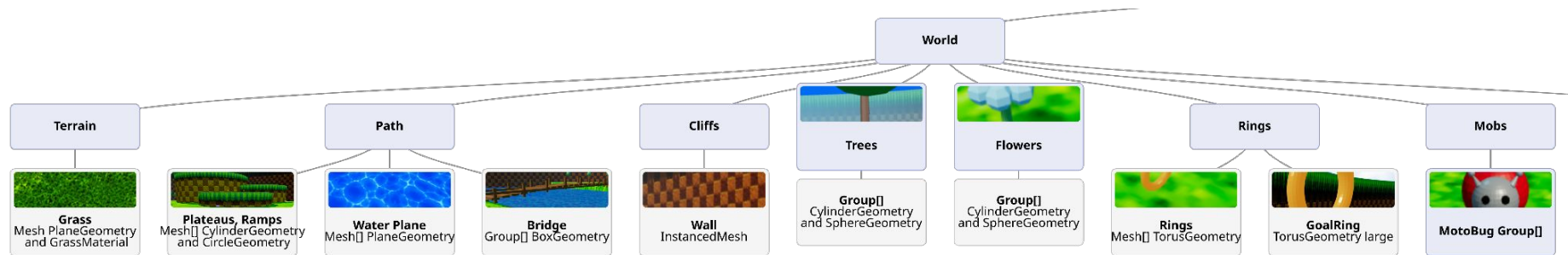
Develop a 3D open world using Three.js, inspired by the Hub World from the Sonic Jam game for the Sega Saturn.



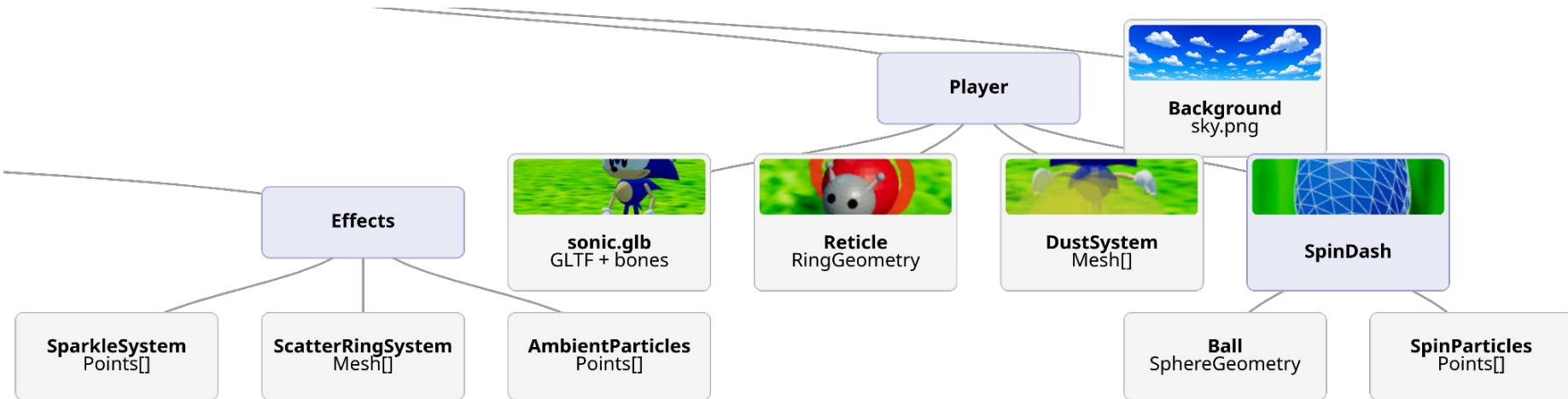
Scene Graph



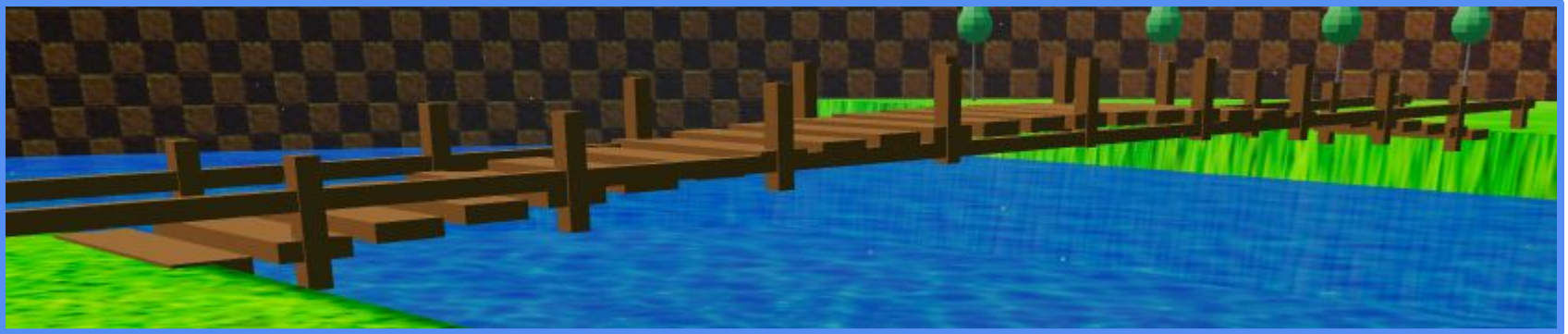
Scene Graph (World)



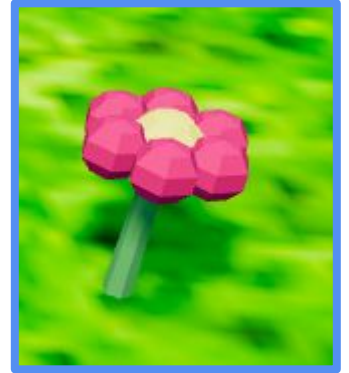
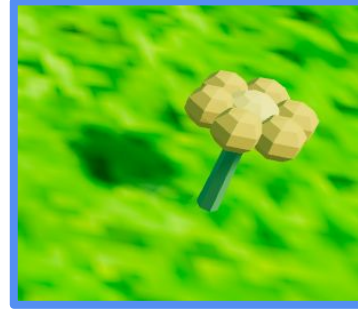
Scene Graph (Player)



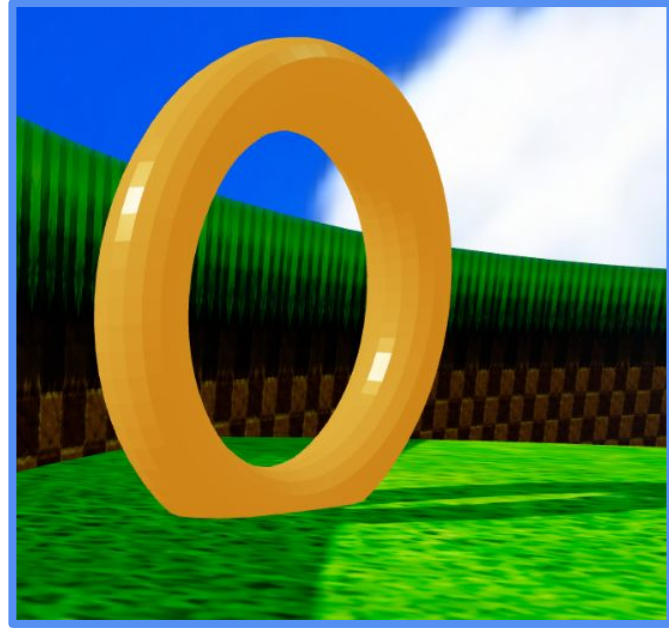
Models (Bridge)



Models (Trees & Flowers)



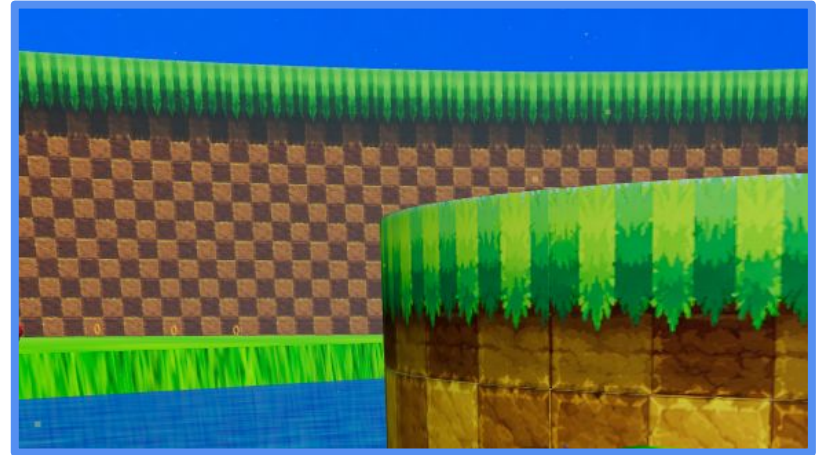
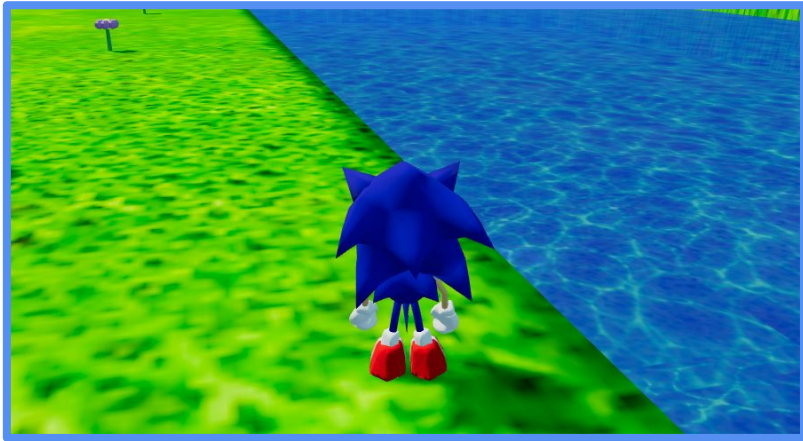
Models (Rings)



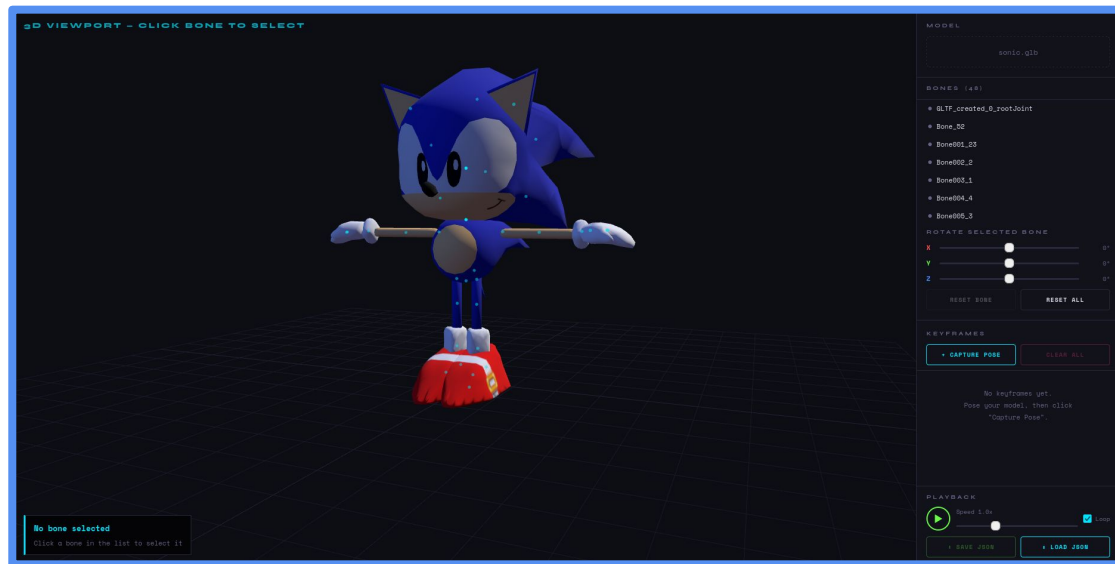
Models (Sonic & Motobug)



Illumination & Materials & Textures



Animation



User Interaction

WASD / Arrows : Move

Space : Jump

Shift / X : Spin Dash

Mouse : Look Around

M : Mute / Unmute

Collect all 50 rings!

As fast as you can



Development

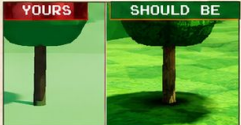
YOUR GAME (CURRENT)



HOW IT SHOULD LOOK (32-BIT STYLE)



WHAT'S WRONG WITH YOUR GAME

1. FLAT LIGHTING	2. LOW-QUALITY TEXTURES	3. NO AMBIENT OCCLUSION	4. UNSATISFYING COLORS	5. OVERLY SMOOTH SHADING
There's no directional lighting or shadows. Everything looks flat and lifeless.	Your textures are too plain and low-res. 32-bit games used bold, detailed textures with clear patterns.	There's no shading in crevices or where objects meet the ground. Things look like they float.	The colors are too pale and washed out. 32-bit games used vibrant, contrasty colors.	The shading is too smooth and modern. 32-bit games used harder edges and simple gradients.
YOURS SHOULD BE 	YOURS SHOULD BE 	YOURS SHOULD BE 	YOURS SHOULD BE 	YOURS SHOULD BE 

HOW TO FIX IT

➤ **ADD DIRECTIONAL LIGHTING**
(Like sunlight from one angle)

➤ **BAKE SHADOWS INTO THE GROUND**
Or use real-time shadows

➤ **USE RETRO TEXTURES**
(Check Sonic Jam, Adventure, or SA1)

➤ **ADD AMBIENT OCCLUSION**
(Especially at the base of objects)

➤ **BOOST COLOR CONTRAST**
(Use brighter colors and deeper shadows)

➤ **USE HARDER SHADING**
(Not soft or blurry lighting)

➤ **LOOK AT REFERENCE GAMES**
(Sonic Jam, SA1, SSB, etc.)

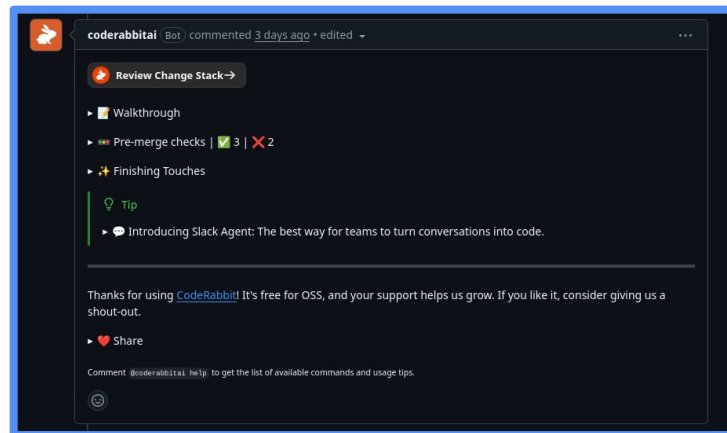
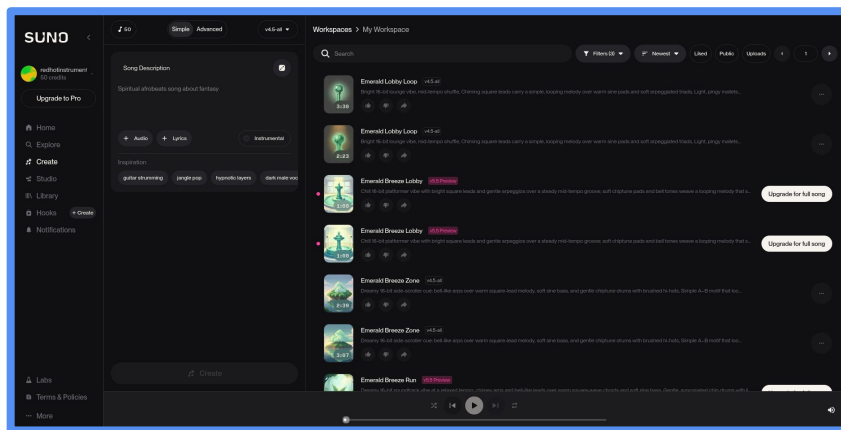
➤ **FOCUS ON READABILITY**
(Important gameplay elements should pop)

GOOD REFERENCES:





AI



Performance



References

Sonic Model:

<https://sketchfab.com/3d-models/sonic-jam-cgi-1c05780f967a475aa92d60cbf937b284>

Sonic Physics:

<https://github.com/ricky-daniel13/HedgePhysicsLite>

Sonic Sound Effects:

<https://github.com/clarkeadg/opensonic-js>

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